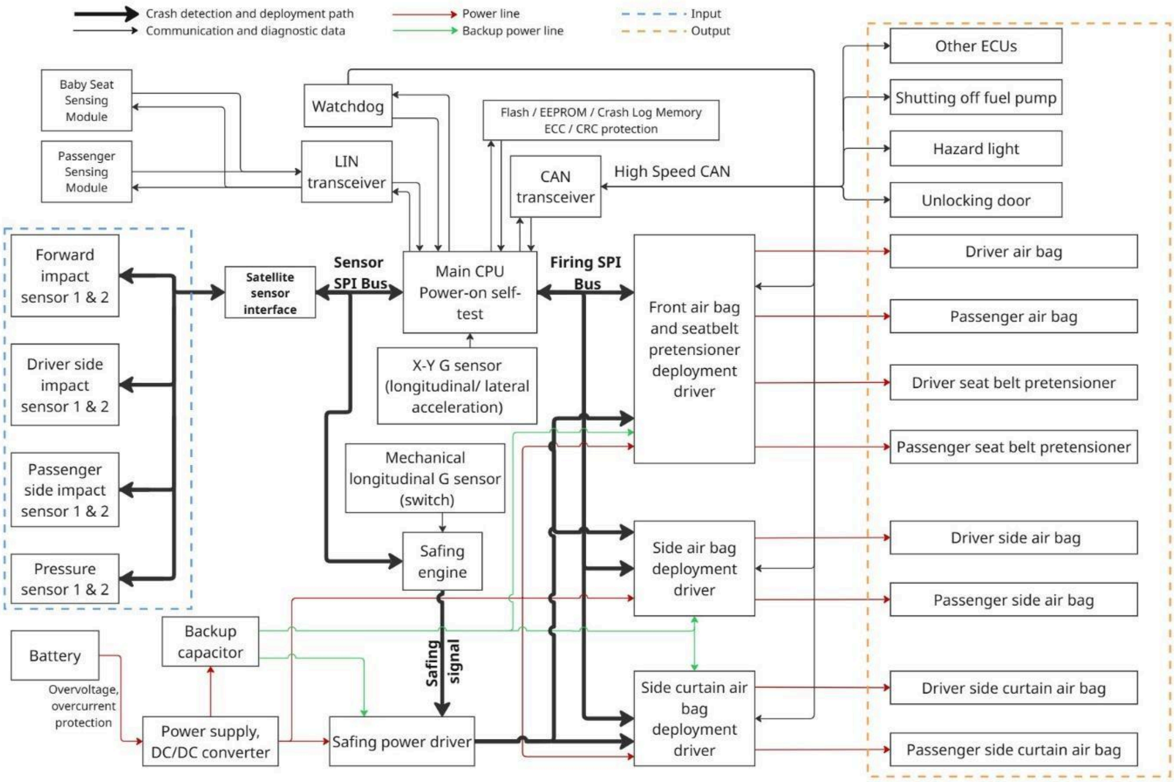
Airbag ECU System Architecture Documentation

Figure 1. Airbag ECU architecture



1. Overview

The airbag Electric Control Unit (ECU) is a safety-critical embedded system responsible for detecting vehicle collisions and deploying appropriate restraint systems such as airbag and seatbelt pretensioners.

This ECU processes data from multiple sensors, evaluates crash conditions in real time, and triggers deployment mechanisms within strict timing constraints while ensuring system safety and redundancy.

2. System Objectives

- Detect frontal, side, and rollover impacts
- Deploy airbags and seatbelt pretensioners within milliseconds
- Ensure high reliability and fault tolerance
- Maintain operation during power interruption (backup energy)
- Communicate with other vehicle ECUs

3. High-level Architecture

The system consists of following main subsystems:

3.1 Sensor Subsystem

Provides crash-related data inputs:

- Forward impact sensors (1&2)
- Driver-side impact sensors (1&2)
- Passenger-side impact sensors (1&2)
- Pressure sensors (1&2)
- X-Y G Sensors (longitudinal & lateral acceleration)
- Mechanical longitudinal G sensor (hardware switch backup)

Interface:

- Sensors communicate via **Sensor SPI Bus**
- Satellite sensors connect through a **Satellite Sensor Interface**

3.2 Processing Unit (Main CPU)

- Performs **Power-On Self-Test (POST)**
- Continuously evaluates sensor data
- Runs crash detection algorithms
- Controls deployment decisions

Connected modules:

- Watchdog (system monitoring)
- Flash/EEPROM (crash logs and configuration)
- CAN & LIN communication interfaces

3.3 Communication Interfaces

- **CAN Transceiver (High-Speed CAN)**
  - Communicates with other ECUs
  - Sends crash status and diagnostics
- **LIN Transceiver**
  - Interfaces with passenger sensing module

3.4 Safety Subsystem

Ensure safe operation and prevents unintended deployment

- **Watchdog**
  - Monitors CPU health and resets on failure
- **Safing Engine**
  - Independent validation of crash conditions
  - Validates crash condition separately from the main CPU
  - Generates the **safing signal**
  - Operates in parallel with the main CPU
- **Mechanical G Sensor**
  - Hardware-related redundant trigger

3.5 Deployment Subsystem

Responsible for triggering restraint devices:

Firing SPI Bus

- Secure communication between CPU and deployment drivers

Deployment Drivers

- Front airbag & seatbelt pretensioners driver
- Side airbag deployment driver
- Side curtain airbag deployment driver

3.6 Actuators (Output)

Triggered during crash events:

- Driver airbag
- Passenger airbag
- Driver seatbelt pretensioner
- Passenger seatbelt pretensioner
- Side airbags (driver and passenger)
- Curtain airbags (driver and passenger)

3.7 Power Supply System

Ensures continuous operation:

- **Battery Input**
  - Primary power source
  - Includes overvoltage/overcurrent protection
- **DC/DC converter**
  - Regulates voltage for ECU components
- **Backup Capacitor**
  - Provides energy during power loss (e.g., crash disconnection)
- **Safing Power Driver**
  - Control energy delivery to deployment circuits

3.8 Airbag System Input/Output Table

Table SEQ Table * ARABIC 1. Airbag system input/output table

		OUTPUT (DEPLOYMENT & ACTIVATION DEVICES			
		Driver & Passenger airbag inflator modules	Driver & Passenger seatbelt pretensioners	Driver-side airbag & driver-side curtain airbag inflator modules	Passenger-side airbag & passenger-side curtain aibag inflator modules
INPUT (SENSOR SIGNAL)	X-Y G Sensor (Longitudinal / Lateral acceleration) in SDM	□	□	□	□
	Mechanical longitudinal G sensor in SDM	□	□	□	□
	Forward-impact sensor	□	□	□	□
	Driver-side impact sensor	□	□	□	□
	Passenger-side impact sensor	□	□	□	□

4. Data Flow Description

4.1 Normal Operation

1. Sensor continuously send data via SPIs
2. The main CPU processes acceleration and pressure signals
3. Crash conditions are evaluated by:
   a. **Main CPU** (software algorithms)
   b. **Safing engine** (independent hardware logic)
4. The main CPU communicates with other vehicle ECUs via:
   a. CAN (High-speed communication)
   b. LIN (e.g., passenger sensing module, baby seat sensing module)

4.2 Crash Event Sequence

1. Impact detected by multiple sensors
2. CPU runs crash algorithm
3. Safing engine validates event independently
4. Deployment is allowed **only when both conditions are satisfied**:
   a. Main CPU confirms the crash event
   b. Safing Engine asserts the safing signal
5. Airbags and pretensioners deploy
6. Crash data stored in Flash memory
7. CAN messages sent to other ECUs (e.g., Fuel cut-off, hazard lights, door power lock)

5. Safety Features

- Dual-path validation (CPU and Safing Engine)
- Redundant sensors (electronic and mechanical)
- Watchdog supervision
- ECC/CRC memory protection

- Backup energy source for deployment feasibility
- Hardware-based safing signal gating

6. Interfaces

Table 2. Airbag ECU interface

Interface	Type	Purpose
Sensor SPI Bus	Input	Sensor data acquisition
Firing SPI Bus	Output	Deployment control
CAN	Bidirectional	Vehicle communication
LIN	Bidirectional	Passenger sensing module

7. Fault Handling

- Watchdog reset on CPU failure
- Fault logging in non-volatile memory
- Diagnostic message via CAN
- Safe state enforcement (no unintended deployment)

8. Compliance Considerations

This architecture is typically designed to meet:

- **ISO 26262** (Functional safety)
- **ASIL-D** requirements for airbag system
- **Automotive EMC** and reliability standards

9. Key Define Principles

- Redundancy
- Deterministic real-time response
- Fail-safe operation
- Independence of safety paths
- Energy availability during crash